

MagicDraw: Haptic-Assisted One-Line Drawing with Shared Control

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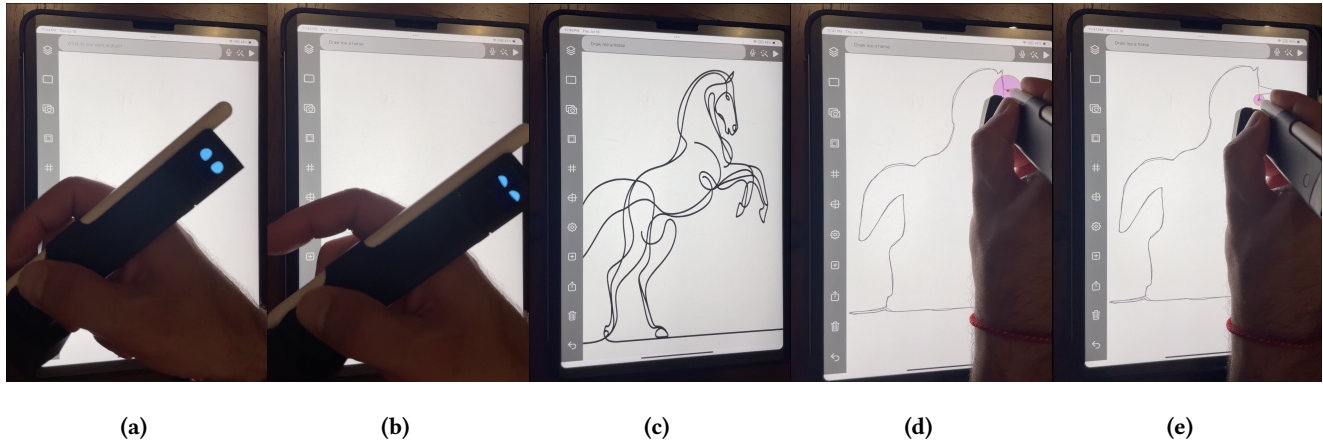


Figure 1: Example interaction with MagicDraw: (a) The user taps to input the prompt, (b) The system sends the user's utterance to GPT-4 Turbo to generate a one-line drawing image, (c) MagicDraw converts the generated image into a series of consecutive points. (d) The MagicPen provides force feedback as the user begins drawing. Applying pressure switches to freehand mode, enlarging uncertainty regions and pausing the path. (e) Lifting the pressure returns the system to fully assisted sketching, with shrinking uncertainty regions as the user resumes tracing.

ABSTRACT

We present MagicDraw, a platform designed for force feedback guidance in one-line drawing. MagicDraw allows users to transition seamlessly between fully assisted sketching and freehand drawing through a control-sharing mechanism. The initial drawing concept is generated based on user input prompts. This platform operates similarly to tracing but introduces two major enhancements. First, MagicDraw provides force feedback guidance, aiding users in maintaining accurate line-following. Second, the system enables dynamic control sharing, allowing users to deviate from the predefined path and engage in creative exploration. We also introduce “exploration region,” where users can perform freehand drawing. In these regions, the predefined path advances outside the boundary, pausing for the user's creative deviations. As the user returns to fully assisted sketching, these regions shrink until the user resumes force feedback-guided tracing. This approach ensures

users can explore creative variations while still receiving structured guidance.

CCS CONCEPTS

• **Human-centered computing** → **Human computer interaction (HCI)**; *Haptic devices*.

KEYWORDS

Assisted Sketching, Force Feedback, Haptics, Shared Control

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1 INTRODUCTION

Various systems and devices have been developed to enhance the physical drawing experience beyond traditional pen-and-paper interaction through force feedback and haptic guidance. RoboSketch [9] links sketches in the physical and virtual worlds, supporting physical sketching of circuits, textile patterns, and markings for woodworking. MusclePlotter [8] generates force feedback through electrical stimulation of the user's muscles, enabling free-roaming

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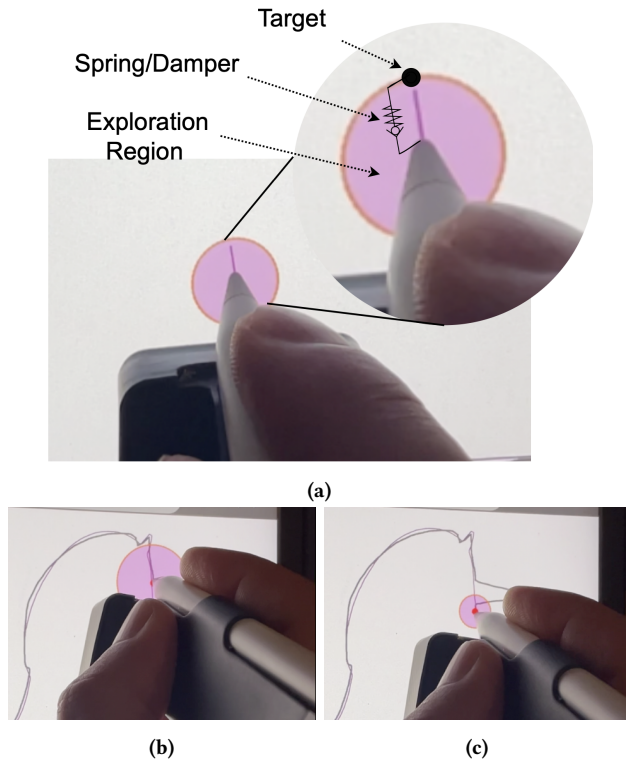


Figure 2: Transition between assisted and freeform drawing (a) The exploration region (purple) surrounds the target point (black dot). The spring/damper model provides force feedback, guiding the user to follow the target. (b) and (c) demonstrate the user receiving haptic feedback to maintain smooth and precise drawing. (b) When the user applies pressure, they can freely deviate within the exploration region; (c) releasing pressure returns them to guided sketching.

sketching, though it has limitations in degrees of freedom and potential muscle adaptation. Additionally, dePENd [11] utilizes ferromagnetic properties of pen ball tips to provide directional force feedback during pen-and-paper interactions, while Phasking on Paper [5] explores shared control between the user and the system using friction-based haptic guides. These approaches focus on seamless transitions between guided and freehand modes.

Several studies have concentrated on supporting users and enhancing their sketching skills in the digital domain. ShadowDraw [7] dynamically updates a shadow image underlying the user's strokes, suggesting object contours and adapting to real-time inputs by blending images from a large database, helping users create more proportioned line drawings. Since early SketchPad [10], digital sketching tools like DressCode [4], iCanDraw [2], DesignScript [1], and Dynamic Brushes [3] have transformed traditional drawing methods, offering intuitive interfaces and features such as dynamic brushes and stroke beautification. MagicDraw extends previous work by focusing on continuity and flow in one-line drawing while allowing expressive freehand drawing as depicted in Figure 1. It emphasizes the smooth transition into and out of shared control

drawing phases, while maintaining simplicity and intuitiveness in modulating control authority for seamless collaboration between the user and the system.

2 MAGICPEN

Our platform uses a 2D force feedback device and add-on for the Apple Pencil called MagicPen. MagicPen builds on our previous ballpoint drive design [5, 6], which we have re-purposed for use on standard-sized tablets like iPads. We reduced the ball size by 50% to 1.27 cm and the overall mechanism size by 30% to 3.2 cm, resulting in a compact 1 cm × 2 cm design resembling a highlighter grip.

The haptic feedback is delivered via a Fluoroelastomer rubber ball, which improves screen friction while minimizing marks. Four coreless gear motors, each capable of 1200 rpm at 3.7V, drive the mechanism. These motors, with custom 6 mm gearheads, can exert a force of up to 0.65 N, ensuring precise and responsive 2D force feedback for an enhanced drawing experience on an iPad.

3 ONE-LINE DRAWING

To evaluate collaborative drawing with force feedback, we focus on the one-line drawing experience. The nature of one-line drawing demands that the artist does not lift the pen from the paper, which serves as an excellent exercise for careful planning and foresight. In addition, engaging in physical sketching and one-line drawing helps in developing fine motor skills, which are essential for precise control over hand movements. This deep immersion in the activity encourages a high level of concentration, which can enhance the overall creative process.

Integrating force feedback into the collaborative one-line drawing experience adds another layer of interaction. Force feedback can guide the artist's hand, providing tactile cues that help maintain the continuous line and potentially improve accuracy and fluidity. This feedback mechanism creates a sense of connection between the artist and the medium, making the drawing process more engaging and intuitive.

4 SHARED CONTROL MECHANISM

This work extends the concept of bringing and control sharing previously discussed in Phasking framework [5]. In the previous approach, a time-based target drawing required users to speed up and follow the line when control sharing occurred. In this work, we introduce a new concept called the “exploration region,” which appears as a circle as shown in Figure 2. Similar to previous approaches, we use pen tip pressure as a means to control the level of shared control.

When the pressure is low (Figure 2c), the MagicPen takes control, and the exploration circle is small. This means the system will pause and apply force feedback to guide the user's hand until it reaches close proximity to the target point, then gradually moves to the next step. Conversely, when the user applies more pressure to the pen tip (Figure 2b), the user takes control, and the system reduces the force feedback accordingly. This allows for freehand drawing, enabling the user to deviate from the predefined line and create new paths.

At the same time, we increase the size of the exploration regions so that the target point can move forward. As the user returns to

the original path and reduces the pen tip pressure, the exploration region become smaller, allowing the user to continue along the guided path.

5 CONCLUSION

In this demo, we present MagicDraw, a novel platform for haptic-assisted sketching and shared control drawing. With MagicDraw, users can seamlessly transition between guided and freehand drawing. The “exploration region” feature allows for creative deviations while maintaining guidance. We believe that shared control drawing will play a crucial role in the future of human-system collaborative drawing.

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